100 DAYS QUALITY MANAGEMENT CHALLENGE • DAY 90

Hindustan Petroleum Corporation Limited (HPCL)

Equipment Reliability and Maintenance Quality: Pump Vibration, Root Cause Analysis & Preventive Maintenance

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1. INTRODUCTION

In continuous process industries such as petroleum refining and processing, high rotating equipment availability and reliability are essential for safe, steady, and profitable operations. Centrifugal pumps handle volatile liquids, hydrocarbons, and utility streams under demanding thermal and pressure conditions.

Unplanned failures or recurring equipment deterioration can result in process upsets, safety hazards, environmental leaks, and substantial production losses. Implementing robust preventive maintenance (PM) practices alongside disciplined root cause failure investigations ensures that mechanical equipment maintains operational integrity throughout its lifecycle.

2. PRACTICAL CASE STUDY (EDUCATIONAL SCENARIO)

Scenario Description

A heavy-duty centrifugal pump deployed in a refinery processing unit develops repeated high-vibration alerts and performance degradation, even after routine maintenance overhauls and bearing replacements.

• **What the Problem Is:** The pump exhibits excessive vibration amplitude during steady-state operation, accompanied by reduced delivery head and minor mechanical seal weeping. The recurring nature of the problem indicates that routine maintenance is addressing symptoms rather than the underlying cause.

• **Why Repeated Vibration Needs Immediate Attention:** Persistent vibration accelerates mechanical fatigue across pump components, damages seal faces, degrades bearing races, loosens foundation anchor bolts, and raises the risk of catastrophic shaft fracture or fluid containment loss.

• **What the Engineer Should Check First:** Prior to invasive dismounting, the engineer should conduct non-intrusive physical and visual checks: foundation anchor bolt tightness, baseplate rigidity, piping strain, oil levels and contamination, overall operating temperature, pump hydraulic operating point (flow rate vs. BEP), and spectrum vibration measurement across drive-end and non-drive-end bearings.

• **How Maintenance Records & Operating Conditions Help:** Reviewing historical log sheets, vibration spectral trends, past work orders, lubrication records, and process parameters (suction pressure, fluid viscosity, operating temperature) helps determine whether the vibration correlates with specific throughput changes, thermal expansion, or recent maintenance interventions.

• **How to Investigate the Possible Root Cause (Evidence-Based Approach):** Avoid assuming a single default cause (e.g., automatically replacing bearings). Perform systematic diagnostic testing: FFT vibration spectrum analysis to distinguish 1X (unbalance), 2X (misalignment), or high-frequency (bearing defect/cavitation) peaks; laser alignment check under cold and hot thermal growth conditions; internal clearance inspection; impeller dynamic balancing; and hydraulic cavitation flow analysis.

• **Corrective Actions:** Implement targeted corrective steps based on diagnostic evidence—such as thermal-growth-compensated laser alignment of the driver-pump shaft, correcting piping stress via flange realignment, rebalancing the impeller assembly to ISO 1940 standards, or adjusting suction pressure to eliminate NPSH margin deficits.

• **Verifying Effectiveness:** Baseline baseline vibration spectra must be captured immediately post-commissioning and re-assessed after 24 hours, 7 days, and 30 days under full thermal load. Operational parameters (bearing temperature, seal integrity, pressure differential) must remain within design limits without trending upward.

3. IMPORTANT QUALITY TOOLS

1. Check Sheet: A structured, real-time data collection form used to log defect counts, inspection metrics, or maintenance observations systematically.
Example: A daily vibration check sheet used by field technicians to record RMS vibration velocity, bearing temperatures, and oil levels across all process pumps.

2. Root Cause Analysis (RCA): A structured problem-solving methodology designed to identify the fundamental physical, human, or systemic causes of an undesirable event rather than merely treating visible symptoms.
Example: Conducting a cross-functional RCA session involving maintenance, operations, and reliability engineers to investigate why Pump P-102 continues to vibrate after alignment.

3. Five Whys: An iterative interrogative technique that explores the cause-and-effect relationships underlying a specific problem by repeatedly asking "Why?" until the root operational or management system failure is uncovered.
Example: Why did the pump vibrate? (Shaft misaligned) → Why misaligned? (Thermal expansion was ignored during cold alignment) → Why ignored? (Alignment protocol lacked hot alignment specifications) → Why lacked? (SOP was not updated after process redesign) → Why not updated? (Lack of management change review procedure).

4. Failure Mode and Effects Analysis (FMEA): A proactive, step-by-step risk assessment tool used to identify potential failure modes, evaluate their causes and severity, and prioritize corrective actions using Risk Priority Numbers (RPN).
Example: Performing an FMEA on centrifugal pump subsystems to score impeller erosion, seal leakage, and shaft misalignment based on Severity, Occurrence, and Detection ratings.

4. KEY ENGINEERING LEARNINGS

- **Equipment Condition:** Vibration and temperature trends serve as reliable indicators of mechanical health; addressing early anomalies prevents catastrophic structural failures.
- **Preventive Maintenance:** PM routines must evolve beyond simple calendar-based component replacement to include precision condition monitoring and alignment standards.
- **Failure Investigation:** Diagnostics must remain strictly evidence-based; jumping to conclusions (such as assuming bearing failure) leads to wasted resources and repeated downtime.
- **Maintenance Records:** Detailed historical logs of operating parameters, work orders, and baseline spectrum data provide critical evidence for diagnosing complex failures.
- **Verification of Corrective Actions:** Corrective interventions are incomplete without post-repair monitoring to verify that equipment operates stably under full working loads.

5. TECHNICAL INTERVIEW QUESTIONS & ANSWERS

Q1: What is the primary difference between Preventive Maintenance (PM) and Corrective Maintenance (CM)?

A: Preventive Maintenance consists of scheduled inspections, lubrication, and servicing to prevent equipment failure. Corrective Maintenance is performed to repair equipment that has already degraded or failed.

Q2: Why should vibration analysis be conducted prior to dismantling a vibrating pump?

A: Pre-dismantling vibration spectrum analysis identifies specific frequencies (e.g., 1X, 2X, vane pass) that pinpoint unbalance, misalignment, or resonance without destroying critical diagnostic evidence.

Q3: How does thermal expansion affect pump-driver shaft alignment in hot fluid applications?

A: Thermal expansion causes machine shafts to shift relative to each other as operating temperatures rise. Alignment performed cold must incorporate manufacturer thermal offsets so shafts align precisely at operating conditions.

Q4: What role do maintenance log sheets play during a root cause failure investigation?

A: Log sheets provide historical trend lines for pressure, temperature, flow, and vibration, allowing engineers to correlate mechanical failure onset with operational changes or past interventions.

Q5: In RCA, what is the danger of stopping the investigation at the physical cause level?

A: Stopping at the physical cause (e.g., "damaged bearing") fixes the symptom but leaves latent systemic causes—such as poor lubrication standards or inadequate training—unaddressed, leading to recurrence.

Q6: How does FMEA help in prioritizing equipment maintenance tasks?

A: FMEA calculates a Risk Priority Number (RPN = Severity × Occurrence × Detection) for each failure mode, enabling engineers to allocate resources to high-risk failure modes first.

Q7: What is piping strain, and how does it contribute to pump vibration?

A: Piping strain occurs when misaligned suction or discharge piping exerts mechanical force on pump nozzles, causing casing deformation, shaft misalignment, bearing overload, and elevated vibration.

Q8: How do you verify whether a corrective maintenance action was successful?

A: By capturing baseline vibration readings, checking operating temperatures, monitoring seal integrity, and confirming stability over extended operation under normal operating loads.

Q9: What is Net Positive Suction Head Available (NPSHa), and how does its deficit cause pump vibration?

A: NPSHa is the absolute suction pressure available above vapor pressure. If NPSHa falls below required NPSHr, fluid vaporizes and collapses (cavitation), causing high-frequency vibration and impeller pitting.

Q10: What is dynamic balancing, and when is it required for pump impellers?

A: Dynamic balancing is the process of adjusting mass distribution around an impeller assembly's rotational axis to minimize centrifugal forces, required whenever impellers are machined, repaired, or replaced.

Q11: How does a Check Sheet differ from a standard quality checklist?

A: A Check Sheet is designed for real-time quantitative/qualitative data collection and defect frequency logging, whereas a checklist is a procedural tool used to confirm task completion.

Q12: What is the significance of a 1X peak in vibration spectrum analysis?

A: A dominant 1X peak (1 times rotational speed) typically signifies mechanical unbalance in rotating components, such as a clogged, worn, or uncorrected impeller.

Q13: Why is evidence-based troubleshooting superior to trial-and-error component replacement?

A: Evidence-based troubleshooting uses data (spectrum analysis, alignment checks, oil samples) to target the actual root cause, avoiding unnecessary component replacement costs and extended downtime.

Q14: How does condition-based maintenance (CBM) improve refinery reliability compared to purely time-based PM?

A: CBM uses real-time diagnostic data to service equipment only when health parameters degrade, eliminating unnecessary overhauls that risk introducing maintenance-induced defects.

Q15: What steps should be taken if a pump continues to vibrate immediately after shaft alignment?

A: Verify baseplate rigidity/soft foot, re-check thermal growth offsets, perform vibration spectral analysis to check for unbalance or resonance, and inspect for external piping forces on casing.

6. REFERENCES & DISCLAIMER

References:

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